

Microalbuminuria and Other Risk Factors of Cardiovascular Disease in Hypertensive Patients in Southern Part of Nigeria

Short Title: Microalbuminuria in hypertensive patients

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ABSTRACT

Microalbuminuria in hypertension has been recognized as a sensitive marker of kidney damage and a predictor for cardiovascular disease and end-stage renal disease. This study was to investigate the level of microalbuminuria in hypertensive patients and its relationship with other risk factors of cardiovascular disease. This was a hospital-based cross-sectional study carried out between October 2022 and November 2023 at the Delta State Central Hospital, Warri. The study included 300 participants consisting of 200 hypertensives and 100 healthy controls. The study used a structured interviewer-administered questionnaire. Data was analyzed with SPSS version 23. The median UACR in the hypertensive was significantly higher than that of the controls. 3.4 (1.8 – 6.3mg/g) vs. 2.0 (1.5 – 3.2mg/g), $p = 0.000$). Thirty (15.0%) of the hypertensive had microalbuminuria compared to none of the controls ($\chi^2 = 8.403$, $p = 0.004$). There was a statistically significant correlation between UACR and waist circumference in the hypertensive population ($r = 0.29$, $p = 0.001$). Similarly, the waist circumference had a positive correlation with UACR ($r = 0.339$, $p = 0.008$) in the control group. Hypertensive kidney damage is present in a significant number of hypertensive patients in this region. Microalbuminuria and UACR were significantly higher in the hypertensive patients than in normotensive controls. In addition, UACR has a significant positive correlation with waist circumference (WC). Periodic assessment for microalbuminuria should be done for hypertensive patients. Also, for individuals with high UACR, closer monitoring of blood pressure is highly recommended.

Keywords: Microalbuminuria, Serum creatinine, Urine albumin creatinine ratio, Hypertension, Waist circumference

INTRODUCTION

Hypertension is one of the main causes of death worldwide and is the leading modifiable risk factor for morbidity and mortality globally. Despite the increasing prevalence of hypertension, a large number of patients are undetected.^{1, 2}

Microalbuminuria in hypertension has been recognized as a sensitive marker of kidney damage and a predictor for cardiovascular disease and end-stage renal disease. It is also believed to be a consequence of renal disease.^{3, 4}

The prevalence of microalbuminuria in hypertensive patients has been reported to range from 4.7% to 58.4%.⁵ It is affected by age, sex, race, severity of the disease and accompanying risk factors.⁶

Microalbuminuria was previously defined as the presence of small quantities of albumin in urine, which is too little to be detected by standard “dip stick” methods.⁷ It is presently defined as “urinary albumin excretion between 30 and 300 mg/day if measured in a 24-hour urine collection, 20 – 200 µg/minute if measured in a timed urine collection or 30 – 300 mg/g if measured with the use of urinary albumin to creatinine ratio in a spot urine collection.”^{8, 9}

Levels of albumin in the urine can be reduced and progression to overt proteinuria can be prevented by aggressive control of blood pressure especially with drugs that block the renin-angiotensin-aldosterone system, and treatment of diabetes.¹⁰

Furthermore, measurement of microalbuminuria, which is a sensitive

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marker of hypertension-induced target organ damage,¹¹ is not routinely done in this environment. Screening for microalbuminuria will enable the identification of patients who would benefit from more aggressive blood pressure control and hence, risk reduction. Thus, findings in this study may justify recommending routine screening for microalbuminuria in hypertensive patients. This study was to investigate the level of microalbuminuria in hypertensive patients and its relationship with other risk factors of cardiovascular disease.

METHODS

Research design and population

This was a hospital-based cross-sectional study conducted between October 2022 and November 2023 at the Delta State Central Hospital, Warri. Three hundred participants consisting of two hundred hypertensive and one hundred normotensive individuals were recruited into the study. The hypertensive and normotensive cases were recruited from the General Outpatient Department and the Cardiology Unit of the Delta State Central Hospital, Warri. The inclusion criteria included participants who were aged eighteen years and above, diagnosed with hypertension (blood pressure $\geq 140/90$ mmHg) and normotensive individuals with blood pressure below 140/90 mmHg. The exclusion criteria for this study were diabetes mellitus, pregnancy, menstruating or breastfeeding women, febrile disease, urinary tract infection, and those participating in vigorous physical activity in the previous 24 hours. Central Hospital Warri is a secondary health facility with 254 beds. They render general and specialized medical and surgical services.

Sample size and sampling procedure

The sample size was calculated from the formula for a cross-sectional study: $n = (Z^2Pq)/d^2$, where n = sample size, Z = standard deviation (1.96), P = prevalence of hypertension (15%), $q = 1-P$ and d = degree

of precision to be used (0.05). Consecutive sampling was used to recruit participants. Written informed consent was obtained from each study participant. The study used a structured interviewer-administered questionnaire, which included identification number, age, gender, weight, height, waist circumference, medical history and laboratory results. Body mass index (BMI) was calculated from the weight (kg) divided by the height (meter) squared. Waist circumference (WC) was measured with a measuring tape at the approximate midpoint between the lower margin of the last palpable rib and the top of the anterior superior iliac crest.

Data collection

Demographic and baseline characteristics

The study used a structured interviewer-administered questionnaire, which included identification number, age, gender, weight, height, waist circumference, alcohol intake and smoking history, medical history and laboratory results. A detailed history was taken including past or current co-morbidities. BP was measured using a mercury sphygmomanometer after the patient had rested for at least ten minutes. The reading at the first appearance of the Korotkoff sound (phase I) was taken as the systolic BP and that at its disappearance (phase V) was taken as the diastolic blood pressure. Hypertension was defined as values 140/90 mmHg and above on two or more different occasions.¹² Height was measured by a stadiometer. Weight was measured using an electronic patient weighing scale. Body mass index (BMI) was calculated from the weight (kg) divided by the height (meter) squared. Obesity was defined as a BMI equal to or greater than 30 kg/m². Waist circumference (WC) was measured with a measuring tape at the approximate midpoint between the lower margin of the last palpable rib and the top of the anterior superior iliac crest.

Blood sample

About five to ten millilitres (5-10 ml) of blood was collected from each participant by venipuncture into a plain bottle and allowed to clot. The specimen was centrifuged at 3000 rpm. The serum was separated from cells using a Pasteur pipette and kept at -20°C until analysis. Morning and evening temperature recordings were taken to monitor temperature of the freezer. Serum creatinine and urine creatinine were assayed on a spectrophotometer using the kinetic modification of the Jaffe procedure. Estimated glomerular filtration rate (eGFR) was predicted from serum creatinine using the Modification of Diet for Renal Disease formula based on age, sex, race and serum creatinine.¹³

Urine sample

About 10 millilitres of random urine was collected into plain bottles and stored at 2 to 8 °C until analysis for urine albumin and creatinine. Urine albumin was assayed on a spectrophotometer using the immunoturbidimetric assay. Urine albumin creatinine ratio was calculated by dividing urine albumin concentration in milligrams by the urine creatinine concentration in grams.

Statistical analysis

Data was analyzed with SPSS version 23. Continuous variables including age, blood pressure measurements, anthropometric variables, and biochemical parameters were tested for normality. Normally distributed continuous variables were summarized as mean, standard deviation, and ranges while skewed variables were summarized as median and interquartile ranges. Categorical variables were summarized using frequencies and percentages.

The differences in means of continuous variables between the hypertensive group and controls were compared using the student T test. Mann Whitney U test was used to compare differences in median between skewed variables. Chi-square test was used for

univariate analysis. The Pearson's correlation coefficient was used to correlate UACR with age, blood pressure measurements, anthropometric variables and eGFR. Statistical significance was set at <0.05.

Ethical Clearance

This study was approved by the ethics and research committee (protocol number: CHW/ECC VOL 1/272) of the Delta State Central Hospital, Warri. Written informed consent was obtained from each study participant and strict confidentiality was observed for participant's data.

RESULTS

A total of 300 subjects including 200 hypertensives and 100 control subjects participated in the study. The age of the hypertensives ranged from 19.0 – 92.0 years with a mean age of 58.2±13.1years. The controls had an age range of 22.0 – 66.0 years. The hypertensives had a peak age group of 50 – 69 years. The hypertensives included 147 (73.5%) females and 53 (26.5%) males while the controls included 53 (53%) females and 47 (47%) males (Table 1).

Three of the hypertensives were underweight, 68 (34.0%) overweight and 72 (36.0%) obese. Two (2.0%) control subject were underweighted, 48 (48.0%) overweight and 32 (32.0%) obese. There was no statistically significant difference in the BMI distribution between the hypertensives and controls ($p = 0.332$). One hundred and sixty-eight (84.0%) hypertensives had central obesity compared to 63 (63.0%) of the controls. The difference in proportion was statistically significant ($p = 0.003$).

Thirty-three (16.5%) of the hypertensives drink alcohol compared to 7 (7.0%) of the controls. The difference in proportion was not statistically significant ($p = 0.104$). None of the hypertensive patients smoke and only 1 (1.0%) of the control smoke cigarettes ($p = 0.723$).

Table 1: Comparison of demographic and biochemical parameters between the hypertensive and control groups

Hypertensive and control groups		Hypertensive	Control	χ^2	P value
Age group					
	18 – 19	0 (0)	0 (0.0)	40.49	<0.001
	20 – 29	3 (1.5)	12 (12.0)		
	30 – 39	5 (2.5)	18 (18.0)		
	40 – 49	44 (22.0)	38 (38.0)		
	50 – 59	53 (26.5)	25 (25.0)		
	60 – 69	53 (26.5)	7 (7.0)		
	≥70	42 (21.0)	0 (0.0)		
Sex					
	Male	53 (26.5)	47 (47.0)	7.200	0.007
	Female	147 (73.5)	53 (53.0)		
BMI					
	Underweight	3 (1.5)	2 (2.0)	3.415	0.332
	Normal	57 (28.5)	18 (18.0)		
	Overweight	68 (34.0)	48 (48.0)		
	Obese	72 (36.0)	32 (32.0)		
Waist Circum.					
	Mean ± SD (cm)	98.78 ± 13.17	91.25 ± 8.14	16.454 ^t	<0.001
Central obesity		168 (84.0)	63 (63.0)	8.721	0.003
Alcohol use		33 (16.5)	7 (7.0)	2.65	0.104
Smoking		0 (0.0)	1 (1.0)	0.126	0.723
UACR					
	Mean ± SD (mg/mmol)	1.02 ± 1.42	0.28 ± 0.16	16.032 ^t	<0.001
	Microalbuminuria	30 (15.0)	0 (0.0)	8.403	0.004
	Normal	170 (85.0)	100 (100.0)		

Abbreviations: BMI: body mass index; UACR: urine albumin creatinine ratio, t= Student's *T*-test

Biochemical parameters

in median was statistically significant ($p = 0.004$).

Renal function

The median (IQR) creatinine level of the hypertensives was 88.4 (70.7 – 114.9) $\mu\text{mol/L}$ while the controls had 88.4 (70.7 – 97.2) $\mu\text{mol/L}$. The difference in median was not statistically significant ($p = 0.234$). The median (IQR) eGFR of the hypertensive was 0.7 (0.6 – 0.9) ml/s/m^2 and the controls had 0.8 (0.7 – 1.1) ml/s/m^2 . The difference

Microalbuminuria

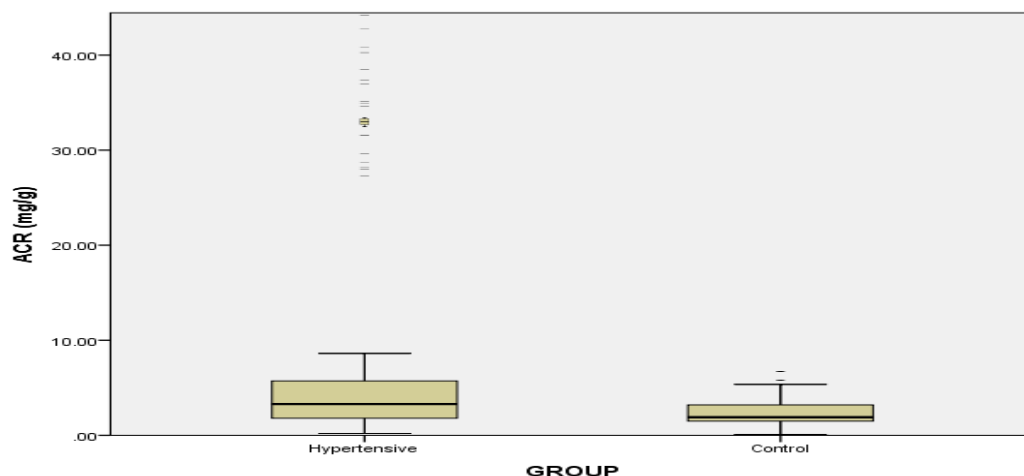
The median UACR in the hypertensives was significantly higher than that of the controls. 0.4 (0.2 – 0.7 mg/mmol) vs. 0.2 (0.2 – 0.4 mg/mmol), $p = 0.000$ (Figure I). Thirty (15.0%) of the hypertensives had microalbuminuria compared to none of the controls and the difference in proportion was statistically significant ($\chi^2 = 8.403$, $p = 0.004$).

Table 2: Age, blood pressure measurements, anthropometric parameters and renal function test

	Hypertensives		Controls		t-test	P value
Age	58.2±13.1	19.0 – 92.0	43±10.7	22.0 – 66.0	7.410	0.001
SBP	169.2±15.6	140.0–220.0	118.3±11.9	90.0–140.0	22.252	0.001
DBP	106.5±9.7	90.0 – 140.0	76.6±7.4	60.0 – 90.0	20.948	0.001
Weight	76.0±18.0	44.0 – 135.0	76.2±9.2	59.0–106.0	-0.067	0.946
Height	1.62±0.07	1.50 – 1.80	1.64±0.09	1.45 – 1.80	-1.265	0.208
BMI	28.9±6.5	16.2 – 51.5	28.6±4.7	18.4 – 41.0	0.338	0.736
WC	98.8±13.2	70.0 – 149.0	91.3±8.1	80.0–129.0	4.056	0.001
Serum Cr*	88.4	70.7 – 114.9	88.4	70.7 – 97.2	Man U	0.234
eGFR*	0.7	0.6 – 0.9	0.8	0.7 – 1.1	Man U	0.004
UACR*	0.4	0.2 – 0.7	0.2	0.2 – 0.4	Man U	0.000

Median, IQR. *Mann Whitney U test.
Abbreviations: eGFR: estimated glomerular filtration rate; WC: waist circumference; Cr: creatinine; BMI: body

mass index; SBP: systolic blood pressure; DBP: diastolic blood pressure; UACR: urine albumin creatinine ratio.

**Figure I: Distribution of albumin creatinine ratio (ACR) in the study population**

Univariate analysis on the association of microalbuminuria with age, sex, BMI, waist circumference, alcohol use and smoking in the hypertensive population.

Table 3 shows the association of microalbuminuria with age, sex, BMI, waist circumference, alcohol use and smoking in the hypertensive population. There were no significant associations with

age group ($p = 0.819$), sex (0.452), BMI (0.218), central obesity (0.806) and smoking. However, alcohol use was

associated with microalbuminuria ($p = 0.001$).

Table 3: Association of microalbuminuria with age, sex, BMI, waist circumference, alcohol use and smoking in the hypertensive population

	Microalbuminuria	Normal UACR	χ^2	P value
Age group				
<40	3(10.0)	5 (2.9)	1.543	0.819
40 – 49	5 (16.7)	39 (22.9)		
50 – 59	5 (16.7)	48 (28.2)		
60 – 69	8 (26.7)	45 (26.5)		
≥ 70	9 (30.0)	33 (19.4)		
Sex				
Male	5 (16.7)	48 (28.2)	0.565	0.452
Female	25 (83.3)	122 (71.8)		
BMI				
Underweight	0 (0.0)	3 (1.8)	4.434	0.218
Normal	15 (50.0)	42 (24.7)		
Overweight	10 (33.3)	58 (34.1)		
Obese	5 (16.7)	67 (39.4)		
Waist Circum.				
Truncal obesity	27 (90.0)	142 (83.5)	0.060	0.806
Normal	3 (10.0)	28 (16.5)		
Alcohol use				
Yes	27 (90.0)	30 (17.5)	34.815	0.000
No	3 (10.0)	140 (82.4)		
Smoking				
Yes	0 (0.0)	0 (0.0)	NA	NA
No	30 (100.0)	170 (100.0)		

Abbreviations: UACR: urine albumin creatinine ratio; BMI: body mass index

Correlations between UACR, age, systolic blood pressure, diastolic blood pressure, anthropometric indices, waist circumference, BMI, serum creatinine and eGFR in the hypertensive subjects and controls

Correlations between UACR, age, systolic blood pressure, diastolic blood pressure, anthropometric indices, waist

circumference, BMI, serum creatinine and eGFR in the hypertensive subjects and controls. There were no statistically significant correlations between UACR and the above parameters in hypertensive population except for waist circumference ($r = 0.29$, $p = 0.001$) but the UACr correlated negatively with diastolic blood pressure ($r = -0.277$, $p = 0.032$) in the control group. Similarly, the waist circumference had a positive correlation with UACR ($r = 0.339$, $p = 0.008$) (Table 4).

Table 4: Correlations between UACR, age, systolic blood pressure, diastolic blood pressure, anthropometric indices, waist circumference, BMI, serum creatinine and eGFR in the hypertensive subjects and controls

	UACR in hypertensives		UACR in Controls	
	R	P value	R	P value
Age	0.016	0.862	-0.086	0.516
SBP	-0.038	0.677	-0.175	0.181
DBP	-0.041	0.660	-0.277	0.032
Weight	-0.071	0.439	0.179	0.171
Height	0.038	0.680	-0.030	0.818
BMI	-0.088	0.339	0.083	0.530
WC	0.29	0.001	0.339	0.008
Serum Cr	0.174	0.058	0.028	0.833
Egfr	-0.046	0.621	0.092	0.482

Abbreviations: UACR: urine albumin creatinine ratio; SBP: systolic blood pressure; DBP: diastolic blood pressure; BMI: body mass index; WC: waist circumference; Cr: creatinine; eGFR: estimated glomerular filtration rate.

DISCUSSION

In this study, results showed that hypertensive subjects had significantly higher levels of microalbuminuria (15%) than the normotensive control subjects. Kim et al, reported a prevalence of 6.6% in a study among adult Korean hypertensive patients.¹⁴ Ardeleanu et al, in a study to assess the prevalence of microalbuminuria in patients with hypertension monitored in primary care, reported a prevalence of 7.1%.¹⁵ In a cross-sectional study by Choi et al, to examine the prevalence and the risk factors of microalbuminuria, a prevalence of 10.1% in 1250 hypertensive subjects was documented.¹⁶ A prevalence of 11% was recorded in a study that included hypertensive Asian Americans.¹⁷ Higher prevalence of 44% and 62.5% have been reported in a one-year prospective multi-

center study of hypertensive patients and a study involving newly diagnosed hypertensive patients respectively.^{18, 19} Prevalence of 22% and 41% recorded in other parts of Nigeria are higher than the prevalence in this study.^{20, 21} The lower prevalence recorded in this study compared with other Nigerian studies and the differences in the prevalence among the different studies may be due to differences in the study population, assay method for measuring urinary albumin levels, severity of hypertension, average age of the study population, racial/ethnic differences and associated co-morbid diseases (for example, diabetes mellitus, kidney diseases).¹⁴

Though Microalbuminuria was not significantly associated with age groups in this study, it was present in all the age groups and the number of subjects with microalbuminuria increased with age. In a study involving hypertensive subjects, microalbuminuria was more prevalent in the higher age group than in the normotensive control subjects.²² Older age was found to be a risk factor for

microalbuminuria in a study consisting of patients with essential hypertension.¹⁸ In addition, older age has been reported as an independent risk factor for microalbuminuria.²³ The proposed mechanism linking microalbuminuria with older age include endothelial dysfunction and arterial sclerosis.²⁴

An elevation of the UACR within the reference limits has been associated with an increased risk of hypertension in the general population.²⁵ In the present study, the median UACR in the hypertensive subjects was significantly higher than in the control subjects. Research involving 1568 non-hypertensive and non-diabetic individuals showed that low levels of UACR below the microalbuminuria range predicted the development of cardiovascular disease.²⁶

Several studies have reported varying results on the association of UACR with obesity measures. In this study, there was a significant association between UACR and central obesity in both the hypertensive and control populations but no significant association was recorded for BMI. Research aimed at investigating the relationship of BMI and WC involving 12672 hypertensive patients showed that BMI was significantly associated with UACR while WC did not correlate.²⁷ In a multinational, observational study of 20828 hypertensive outpatients, abnormal WC, but not BMI was associated with microalbuminuria.²⁸ Ren et al, reported that BMI, WC and body fat content were significantly correlated with UACR.²⁹ The suggested mechanism linking obesity with microalbuminuria is endothelial dysfunction which is a consequence of several circulating inflammatory mediators produced in adipose tissue.³⁰

The limitation of this present study was the single measurement of UACR because the rate of albumin loss is affected by a variety of physiological factors (e.g., exercise, posture, time of day), and the variability is such that repeated measurement is more reliable.

CONCLUSION

The prevalence of microalbuminuria was 18% in the hypertensive group, indicating that hypertensive kidney damage is present in a significant number of hypertensive patients in this region. Microalbuminuria and UACR were significantly higher in the hypertensive patients than in normotensive controls. In addition, UACR has a significant positive correlation with WC. Periodic assessment for microalbuminuria should be done for hypertensive patients. Also, for individuals with high UACR, closer monitoring of blood pressure is highly recommended.

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CONFLICTS OF INTEREST

The authors declare that they have no conflict of interest concerning this article.

CONTRIBUTORS

AOE, DFE and EBG made contributions to the conception, design, experimentation, acquisition and interpretation of data and writing of manuscript. AOE and HES made substantial contributions in interpretation of data and revising the manuscript. All authors read and approved the final manuscript.

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